Secure Examination Question Paper Protection Using Hybrid Cryptography

A Thesis (Major Project) submitted in partial fulfilment of the requirements

For the Degree of

Bachelor of Technology

In

Computer Science and Engineering

Submitted by

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We, the students of Computer Science & Engineering, Barak Valley Engineering College, declare that the work entitled **"Secure Examination Question Paper Protection Using Hybrid Cryptography"** has been successfully completed under the guidance of **Mr. Swmdwn Basumatary**, Assistant Professor, Department of Computer Science & Engineering, Barak Valley Engineering College.

The report has been prepared without resorting to plagiarism. We have adhered to all principles of academic honesty and integrity. No falsified or fabricated data have been presented in the report. Further, the matter embodied in the project report has not been submitted previously by anybody for the award of any degree or diploma to any university.

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Abstract

Examination question papers are among the most sensitive assets of an educational institution. A leak before the scheduled examination, an undetected modification, or a premature release can invalidate an entire examination cycle. Conventional digital systems typically store papers on a central server where a breached server or a malicious administrator can expose plaintext content. This project proposes the design, implementation and comparative performance evaluation of two hybrid-cryptographic architectures for protecting question papers during storage and distribution. Both suites encrypt papers with AES-256-GCM and sign the protected package with Ed25519. They differ in how the AES data-encryption key is protected: Suite A uses RSA-OAEP with 2048-bit and 3072-bit keys, whereas Suite B uses ephemeral X25519 key agreement with HKDF-based key derivation and authenticated key wrapping. The suites will be compared on key-generation time, key-protection and recovery time, encryption and decryption latency across document sizes, throughput, storage overhead, and memory and CPU utilization. Functional security tests will check that modified, incorrectly signed, or incomplete packages are rejected. The current stage of the project covers the cryptographic layer only; a Role-Based Access Control (RBAC) and Temporal RBAC (TRBAC) system for managing permissions and examination-time restrictions is proposed as future work.

`Keywords:` Hybrid Cryptography, AES-256-GCM, RSA-OAEP, X25519, HKDF, Ed25519, Examination Security, Digital Signatures

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Chapter 1

Introduction

1.1 Introduction to the Topic

The security of digital information has become an important concern for educational institutions as they increasingly adopt digital systems for preparing, storing, and distributing examination question papers. These documents contain confidential information whose premature disclosure or unauthorized modification can compromise the fairness and integrity of an examination. Such incidents may result in examination cancellations, financial losses, administrative complications, and a loss of trust among students and educational authorities. Consequently, reliable mechanisms are required to protect examination question papers against unauthorized disclosure and tampering.

Conventional document management systems may rely heavily on centralized storage and application-level security mechanisms. Although these systems simplify document handling, they may expose sensitive information if an attacker gains access to the storage infrastructure or if the application is compromised. Cryptographic protection provides an additional layer of security by ensuring that stored documents remain unreadable without the appropriate decryption key and that unauthorized modifications can be detected.

This project focuses on the implementation and comparative performance evaluation of two hybrid cryptographic approaches for protecting online examination question papers. Both approaches use the Advanced Encryption Standard in Galois/Counter Mode (AES-256-GCM) to encrypt question-paper contents. Since symmetric encryption is computationally efficient for large documents, AES-256-GCM serves as the common encryption mechanism in both approaches. It also provides authenticated encryption, enabling the detection of modifications to the encrypted content.

The first approach, referred to as Suite A, uses RSA with Optimal Asymmetric Encryption Padding (RSA-OAEP) and 2048-bit and 3072-bit key configurations to protect the AES data-encryption key. The second approach, referred to as Suite B, uses ephemeral X25519 key agreement, followed by the HMAC-based Key Derivation Function (HKDF), to derive key material for protecting the data-encryption key. These approaches will be evaluated under documented experimental conditions to examine their performance and resource requirements.

Digital signatures using Ed25519 will provide an additional mechanism for verifying the authenticity of protected examination packages and detecting unauthorized

modifications to signed data. Where browser-side processing is implemented, signature verification and decryption will be performed in the client environment to investigate the extent to which plaintext exposure on the application server can be reduced.

The comparative evaluation will consider parameters such as key-generation time, key-protection and recovery time, encryption and decryption latency, throughput, storage overhead, and computational resource utilization. Functional security tests will examine successful encryption and decryption, signature verification, and the rejection of modified or incorrectly signed packages. The results will be used to assess the practical performance and limitations of the two cryptographic approaches.

1.2 Background

Hybrid cryptography combines symmetric and asymmetric techniques to balance computational efficiency and key management. Symmetric encryption is suited to protecting large amounts of data, whereas asymmetric mechanisms support key establishment, key protection, and digital signatures. This combination fits examination question papers, which must remain confidential while being stored and transferred between system components. The AES key that encrypts a paper must itself be protected, so that access to the encrypted document alone does not reveal its contents. The two suites studied here take different approaches to this problem, described in Chapter 3.

The security of a hybrid system depends not only on the chosen algorithms but also on their implementation, on key management, and on how plaintext is handled. Encryption does not prevent disclosure if decryption keys are improperly managed or if plaintext is exposed during processing.

1.3 Motivation

Examination question papers are high-value targets: a single leak can force a re-examination, and the damage to an institution's credibility is hard to undo. In digital workflows the risk is concentrated where papers are stored and processed, because anyone with access to the storage layer, or an attacker who compromises it, may see the content in plaintext. Application-level access restrictions alone do not remove this risk. Cryptographic protection of the paper itself does: if the stored package is encrypted, authenticated and signed, access to the storage reveals nothing readable and tampering can be detected.

This project therefore aims to establish a practical cryptographic foundation for protecting question papers and to answer a concrete design question: how do RSA-OAEP-based and X25519-based key protection compare in cost and overhead when

every other component is held constant? The studies reviewed in Chapter 2 do not provide a controlled comparison of RSA-based and ECC-based suites on identical payloads. A tested cryptographic layer can also serve as the base for later role-based and time-based access control.

1.4 Objectives

The primary objectives of the project are:

- To implement two hybrid cryptographic suites for examination question-paper protection that share AES-256-GCM and Ed25519 and differ in key protection: RSA-OAEP (Suite A) and X25519 with HKDF (Suite B).
- To implement digital-signature generation and verification to establish the authenticity of protected examination packages and detect unauthorized modifications.
- To compare RSA-2048, RSA-3072, and the selected X25519-based approach using key-generation time, key-protection and recovery time, encryption and decryption latency, throughput, storage overhead, and memory utilization.
- To test the functional correctness of the cryptographic mechanisms, including successful decryption, rejection of modified ciphertext or authenticated metadata, and rejection of invalid signatures.
- To investigate browser-side signature verification and in-memory decryption as a means of reducing unnecessary exposure of plaintext examination papers on the application server.
- To identify the limitations of both suites and the assumptions on which their security depends.

1.5 Scope of the Project

The scope of this project is the cryptographic protection of online examination question papers. It covers AES-256-GCM encryption; DEK protection using RSA-OAEP with 2048-bit and 3072-bit keys (Suite A) and ephemeral X25519 key agreement with HKDF and authenticated key wrapping (Suite B); Ed25519 signature generation and verification; and browser-side verification and decryption. The three configurations will be compared in controlled experiments measuring execution time, key-generation overhead, throughput, storage requirements, and memory and CPU utilization, and functional

security tests will examine successful decryption, signature verification, and detection of modified packages.

The following are outside the scope of the current stage: role-based and temporal access control, user authentication, audit logging, and a complete examination-management workflow. Protection against compromised client devices, theft of private keys, and capture of displayed content is also outside the scope. Access control is identified as future work in Chapter 5.

1.6 Organization of the Thesis

The remainder of this report is organized into the following chapters. Chapter 2 presents the literature survey and reviews relevant existing research and systems. Chapter 3 describes the methodology, architecture, algorithms, modules, and implementation approach. Chapter 4 presents the results and discussion. Chapter 5 presents the conclusion and future work.

Chapter 2

Literature Survey

2.1 Hybrid Cryptography

Symmetric ciphers such as AES are fast and suited to bulk data, but they require both parties to share a secret key in advance. Asymmetric cryptography removes this need but is slow for large data. Hybrid encryption combines them: the data is encrypted with a random symmetric key, and that key is protected or derived with public-key operations. This key-encapsulation and data-encapsulation structure has a formal security analysis [1]. The techniques used in this project are standardized: RSA-OAEP is specified in PKCS#1 [2], and Diffie–Hellman-based hybrid public-key encryption in RFC 9180 [3]. AES in Galois/Counter Mode (GCM) additionally provides integrity through an authentication tag and requires a unique nonce for every encryption under the same key [4].

2.2 AES and ECC-Based Hybrid Schemes

Koupaei [5] proposes a hybrid scheme combining AES for data encryption and ECC for key exchange. The scheme generates a random AES session key, encrypts the data, and encrypts the session key under each recipient's ECC public key. It lists key management, authentication, hybrid key agreement, and testing as phases, and argues for smaller keys and lower overhead than RSA. The paper also discusses drawbacks such as implementation complexity, key-management difficulty, and compatibility. Its experimental section presents qualitative score charts and timing plots, but gives limited detail on the implementation, hardware, and measurement method.

Kumar and Kumar [6] propose an AES–ECC approach for cloud storage in which the AES key is derived using ECC. They evaluate encryption and decryption time on image datasets of 2823, 3482, and 5890 images, and report avalanche-effect, energy-consumption, and key-size comparisons against DES, AES, Blowfish, and a base paper. They report that the hybrid scheme encrypts and decrypts faster and has a higher avalanche effect (above 50%). Their review also notes that authentication and key handling remain open concerns in cloud settings.

2.3 RSA versus ECC

Both papers [5, 6] argue that ECC offers comparable security with much smaller keys. This agrees with standard guidance: for roughly 128-bit security, NIST SP 800-57 [7] associates RSA with a 3072-bit modulus and elliptic curves with about 256-bit keys, whereas RSA-2048 corresponds to roughly 112-bit security. Curve25519 targets approximately the 128-bit level [8, 9], so RSA-3072 is the closer security-level match to X25519, and RSA-2048 is included as a widely deployed baseline. Curve25519 public keys are 32 bytes [8], whereas an RSA-3072 modulus is 384 bytes, so ECC reduces transmission and storage overhead and speeds up key generation.

Performance, however, does not follow key size alone. Gura et al. [10] found that the advantage of ECC over RSA grows with key size and on processors with smaller word sizes. On an 8 MHz ATmega128 microcontroller, however, an RSA-1024 public-key operation with exponent $2^{16}+1$ (0.43 s) was faster than a 160-bit ECC point multiplication (0.81 s). RSA public-key operations such as encryption and signature verification are fast, and RSA has very wide support. The relative cost of RSA and ECC in a complete hybrid workflow running in a browser is therefore an empirical question that depends on the implementation and operation mix, which motivates direct measurement in this project.

2.4 Access Control in Secure Systems

Role-Based Access Control (RBAC) [11] grants permissions to roles rather than individuals, which simplifies administration and supports least privilege. Temporal RBAC (TRBAC) [12] extends this with temporal constraints, such as the periodic enabling and disabling of roles. For examinations, time is a first-class security attribute: a paper must be unreadable before its release window, even to otherwise authorized roles. Combining TRBAC with cryptographic enforcement, rather than relying on server checks alone, strengthens this guarantee. Access control is outside the scope of the current stage and is reviewed here only to motivate the future work in Chapter 5.

2.5 Review of Related Work and Gaps

From this review we identify the following gaps. (i) Among the studies reviewed, RSA-versus-ECC comparisons are made either for single primitives on embedded hardware [10] or between ECC-based schemes and symmetric ciphers [6]; none compares complete RSA-based and ECC-based hybrid packages (key protection, authenticated encryption, and signature) on identical payloads with a reproducible methodology. (ii) Key establishment is often presented without explicit authentication, leaving

Table 2.1: Summary of reviewed work and limitations

Reference	Technique	Limitation relevant to this project
Koupaei [5]	AES–ECC hybrid encryption; ECC protects the AES session key	Qualitative scores; limited reproducible methodology; no explicit authentication or access-control design
Kumar and Kumar [6]	AES–ECC for cloud storage, evaluated using execution time, avalanche effect, and energy consumption on image data	Compared mainly against symmetric ciphers; no signature-based authentication; no controlled RSA-versus-ECC comparison on identical payloads; no temporal access control
Gura et al. [10]	Primitive-level comparison of RSA and ECC on 8-bit microcontrollers	Single operations on embedded hardware; no complete hybrid workflow (AES-GCM, signatures) and no browser-side measurement

impersonation and man-in-the-middle risks. (iii) Client-side verification and in-memory decryption are seldom studied together with server-side measurements. (iv) Cryptography is rarely integrated with temporal and role-based access control for a realistic examination workflow. This project addresses gaps (i) to (iii); gap (iv) is left for future work.

2.6 Problem Statement

Examination question papers must be protected against unauthorized disclosure and modification while they are stored and transferred. If an attacker gains access to stored documents that are not cryptographically protected, the confidentiality and integrity of the examination may be compromised. Hybrid cryptography offers established mechanisms for this protection, but the performance and resource cost of RSA-based and ECC-based realizations vary with the algorithms, key sizes, and workflows used, and the reviewed literature offers no reproducible comparison of complete RSA-based and ECC-based hybrid packages on identical payloads.

This project therefore implements two hybrid suites, compares them under documented experimental conditions, and tests the correctness of encryption, key protection, signature verification, and detection of modified packages.

Chapter 3

Detailed Methodology

3.1 Overview of the Proposed System

This project implements and evaluates two hybrid cryptographic suites for protecting online examination question papers. Both suites encrypt the paper with AES-256-GCM and sign the protected package with Ed25519. They differ only in how the AES data-encryption key (DEK) is protected: Suite A uses RSA-OAEP with 2048-bit and 3072-bit keys, and Suite B uses X25519 key agreement, HKDF-based key derivation, and authenticated key wrapping.

The methodology has two workflows: protection and storage of a question paper (Phase I), and verification and decryption of a stored package (Phase II). Both suites are evaluated through controlled experiments on computational overhead, resource requirements, and functional security.

3.2 Proposed System Architecture

3.2.1 Cryptographic Components

AES-256-GCM

The Advanced Encryption Standard (AES) is a symmetric encryption algorithm that uses the same secret key for encryption and decryption. AES-256-GCM is selected to protect examination question papers because it combines confidentiality with authenticated encryption.

For each encryption operation, a cryptographically secure random number generator will generate a 256-bit DEK. A unique nonce will be used with the key for each encryption operation. AES-GCM produces the ciphertext and an authentication tag, which enables the detection of unauthorized modifications to the encrypted data and authenticated inputs.

The ciphertext, nonce, authentication tag, and required metadata will form part of the protected examination package. AES-256-GCM will remain common to both suites to provide a consistent basis for comparing their asymmetric key-protection mechanisms.

RSA-OAEP

RSA is an asymmetric cryptographic algorithm that uses a public-key and private-key pair. RSA-OAEP is used to protect small amounts of data, such as the AES DEK, rather than encrypting the entire examination document.

Suite A will evaluate RSA-OAEP with 2048-bit and 3072-bit RSA key configurations. The public key will be used to protect the DEK, while the corresponding private-key operation will recover it when decryption is required. Private-key handling will be defined by the implemented key-management procedure.

The evaluation will measure key-generation time, key-protection and recovery time, and the contribution of RSA operations to the overall hybrid encryption workflow.

X25519 and HKDF

Suite B will use X25519, an elliptic-curve Diffie–Hellman key-agreement mechanism, to establish shared secret material. HKDF will derive suitable cryptographic key material from the shared secret.

The derived material will be used by an authenticated key-wrapping mechanism to protect the AES DEK. The implementation will specify the participating key pairs, derivation context, required protocol parameters, and authentication procedure.

Unlike RSA-OAEP, X25519 does not directly encrypt the AES key. It establishes shared secret material from which the necessary key-encryption material is derived. This distinction will be reflected in the implementation and performance measurements.

Ed25519 Digital Signatures

Ed25519 is an elliptic-curve digital signature scheme that uses a private signing key to generate signatures and a corresponding public key to verify them.

The system will use Ed25519 to sign the defined contents of a protected examination package, including the ciphertext and relevant metadata. The receiving component will verify the signature using a trusted public key before accepting the package for decryption.

Digital signatures complement AES-GCM authentication. Ed25519 verifies the authenticity of the signed package, while the AES-GCM authentication tag detects modifications to the encrypted data during decryption.

The same signature mechanism will be used for both suites to keep the principal comparison focused on their respective key-protection approaches.

3.2.2 Phase I: Question Paper Encryption and Secure Storage

Figure 3.1 illustrates the process of protecting an examination question paper before storage.

Phase I: Question Paper Protection

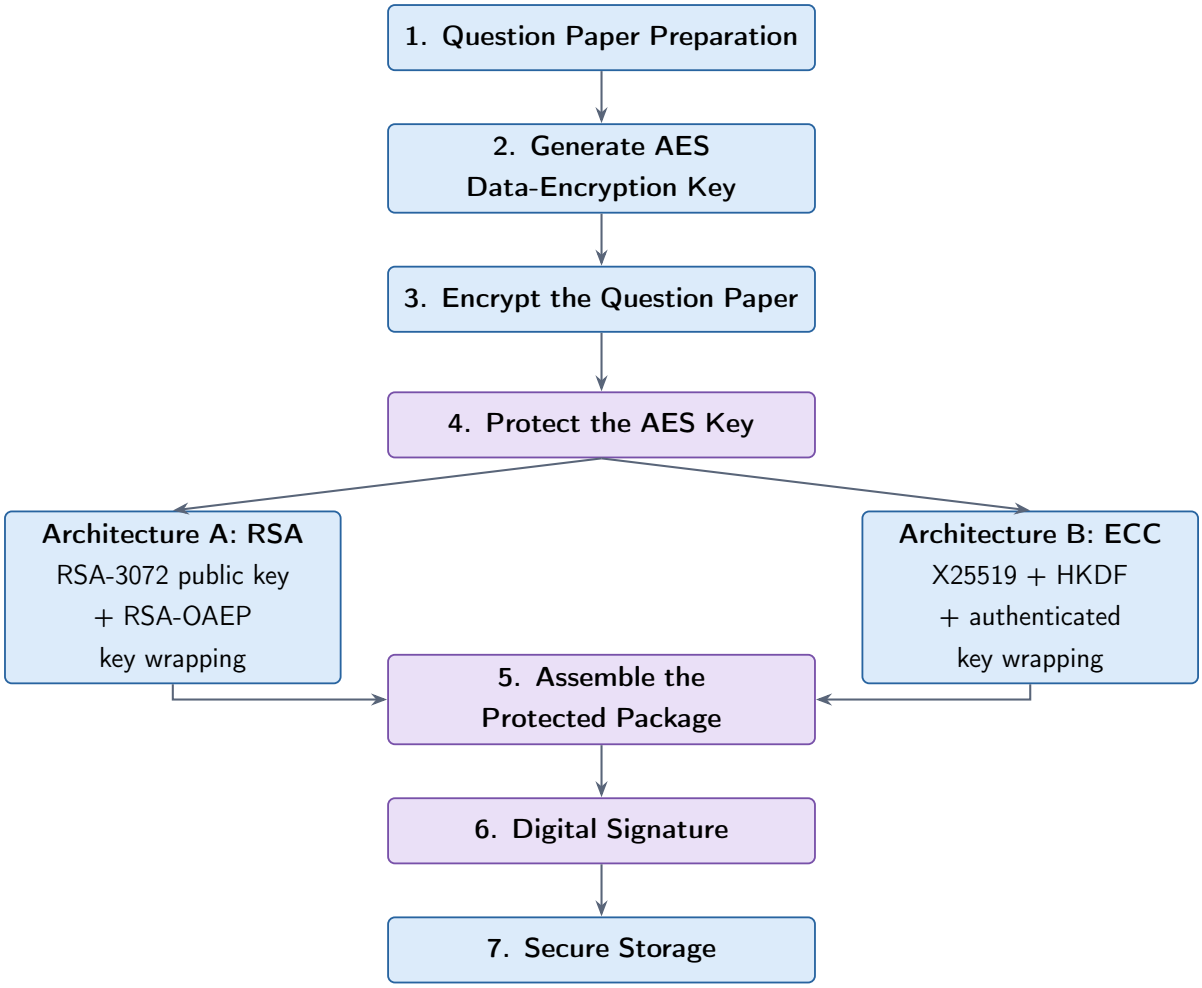


Figure 3.1: Phase I: Question Paper Encryption and Secure Storage.

The process begins with the submission of a question-paper document. The system associates the document with the required metadata, such as an examination identifier and document version.

A random AES-256 DEK is generated and used to encrypt the document through AES-256-GCM. The resulting ciphertext, nonce, and authentication tag are retained as part of the protected package.

The DEK is then protected using one of the two approaches:

- **Suite A:** RSA-OAEP protects the DEK using the designated RSA public key.
- **Suite B:** X25519 and HKDF establish the key material required by the authenticated key-wrapping mechanism.

The encrypted document, protected key material, and associated metadata are assembled into a package. Ed25519 is used to sign the defined package contents. The

protected package is then stored without requiring the ordinary storage layer to retain the question paper as plaintext.

3.2.3 Phase II: Cryptographic Verification and Decryption

Figure 3.2 presents the process of retrieving and decrypting a protected examination package.

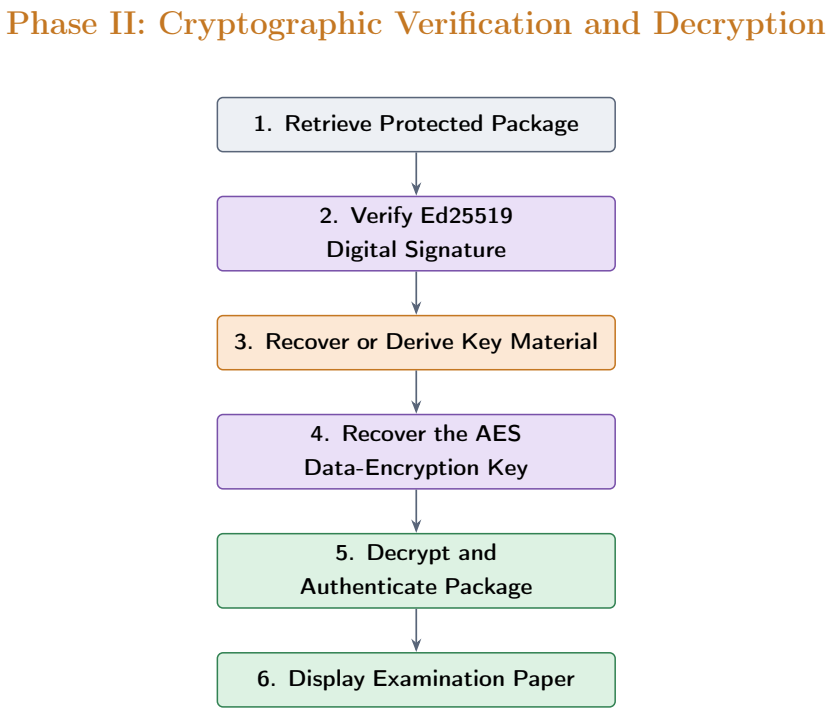


Figure 3.2: Phase II: Cryptographic Verification and Decryption.

The protected package is retrieved from storage and provided to the component responsible for verification and decryption. Ed25519 signature verification is performed using the trusted public key. If verification fails, the package is rejected.

Following successful verification, the required key material is recovered or derived using the procedure corresponding to the selected suite. For Suite A, this involves RSA-OAEP decryption of the protected DEK. For Suite B, it involves the specified X25519 key-agreement, HKDF, and authenticated key-wrapping procedure.

The recovered DEK is used to decrypt the ciphertext through AES-256-GCM. If authentication fails, the plaintext is not accepted. Where browser-side decryption is implemented, the decrypted document is handled in browser memory to reduce unnecessary exposure of plaintext on the application server.

This workflow evaluates the correctness of package verification, key recovery, and decryption. Browser-side processing can reduce server-side plaintext exposure, but it does not eliminate risks associated with compromised client devices or the capture of displayed content.

3.3 Comparative Performance Evaluation

3.3.1 Experimental Configurations

The research will evaluate three configurations:

- RSA-2048 with RSA-OAEP for DEK protection.
- RSA-3072 with RSA-OAEP for DEK protection.
- X25519-based key agreement with HKDF and authenticated key wrapping.

All configurations will use AES-256-GCM for document encryption and Ed25519 for digital signatures. Keeping these components common provides a consistent basis for investigating the performance differences associated with the asymmetric key-management procedures.

The experiments will use a documented hardware and software environment. Where applicable, the same test documents, document sizes, implementation environment, and measurement procedures will be used across configurations.

3.3.2 Performance Metrics

The following metrics will be measured:

- **Key-generation time:** time to generate the RSA or X25519 key pairs.
- **Key-protection and recovery time:** time to protect the DEK and to recover or derive it (RSA-OAEP encryption and decryption; X25519 agreement, HKDF, and unwrapping).
- **Encryption and decryption latency:** time for AES-256-GCM alone and for the complete hybrid workflows, including signing and verification.
- **Throughput:** document data processed per unit time (MB/s).
- **Memory utilization:** memory consumed during key generation and cryptographic operations.
- **CPU utilization:** processor load during the same operations.
- **Storage overhead:** additional bytes for wrapped key material, nonce, tag, signature, and metadata.

Individual operations are timed separately from the end-to-end workflow, to show how much the asymmetric operations contribute to total cost.

3.3.3 Experimental Procedure

Experiments will use test documents of several sizes (for example 100 KB, 1 MB, 10 MB, and 50 MB). Each operation will be run after warm-up runs and repeated 30 times per configuration and size. The mean, median, and standard deviation will be reported, and results will be presented in comparative tables and graphs. Throughput, memory and CPU utilization, and storage overhead will be compared in the same way. Performance results will be interpreted alongside the functional security tests; faster execution is not evidence of stronger security.

3.4 Security Objectives and Testing

The implementation will be evaluated against the following objectives.

3.4.1 Confidentiality

AES-256-GCM will encrypt examination question papers, while the selected asymmetric mechanism will protect the DEK. The effectiveness of this protection depends on correct key management and the prevention of unauthorized access to decryption capabilities.

3.4.2 Integrity and Authenticity

AES-GCM authentication will detect modifications to encrypted data and authenticated inputs. Ed25519 signature verification will establish whether a package matches the signed contents and was signed by the holder of the corresponding private key, provided that the verification public key is trusted.

3.4.3 Functional Security Testing

Testing will cover the following cases:

- Successful encryption, decryption, signing, and verification of valid packages in both suites.
- Rejection of packages whose signed data has been modified.
- Rejection of packages with an invalid signature or a signature from an untrusted key.
- Detection, through AES-GCM authentication, of modified ciphertext, nonce, tag, or authenticated metadata (for example an altered exam identifier or version).
- Failure of key recovery or decryption when the wrong recipient key is used.

- Rejection of packages with missing or malformed cryptographic components.

These tests will assess the correctness of the implemented cryptographic mechanisms. They will not be treated as a comprehensive assessment of every possible attack against the wider examination environment.

Chapter 4

Results and Discussion

4.1 Implementation Overview

4.2 Experimental Results

4.3 Feature Validation

4.4 Discussion

4.5 Testing

Chapter 5

Conclusion

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